from scapy.all import

```
from scapy.all import sniff, IP, TCP
from collections import defaultdict
from datetime import datetime
```

Track packets and destination ports

```
packet_count = defaultdict(int)
scanned_ports = defaultdict(set)
alerted = set()

def detect_intrusion(packet):
    if IP not in packet:
        return

    src_ip = packet[IP].src
    dst_ip = packet[IP].dst

    packet_count[src_ip] += 1
```

Check for possible TCP port scanning

```
    if TCP in packet:
        port = packet[TCP].dport
        scanned_ports[src_ip].add(port)

        if len(scanned_ports[src_ip]) >= 10:
            alert_key = (src_ip, "Port Scan")

            if alert_key not in alerted:
                print("\n[ALERT] Possible Port
Scan!")
                print("Source IP:", src_ip)
                print("Unique ports:",
len(scanned_ports[src_ip]))
                alerted.add(alert_key)
```

Simple high-traffic threshold for testing

```
    if packet_count[src_ip] >= 100:
        alert_key = (src_ip, "High Traffic")
```

```
        if alert_key not in alerted:
            print("\n[ALERT] High Traffic
Activity!")
            print("Source IP:", src_ip)
            print("Packet count:",
packet_count[src_ip])
            alerted.add(alert_key)
```

Display packet information

```
    print(
        datetime.now().strftime("%H:%M:
%S"),
        src_ip, "->", dst_ip
    )

print("NIDS started. Monitoring authorized
traffic...")

sniff(
```

```
    filter="ip",
    prn=detect_intrusion,
    store=False
)
```